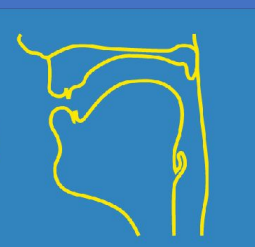


Vowel Allophony Improves Maximum-Likelihood



Classification of Warlpiri Consonants



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Background: Australian Languages

Many consonant places of articulation (POAs)

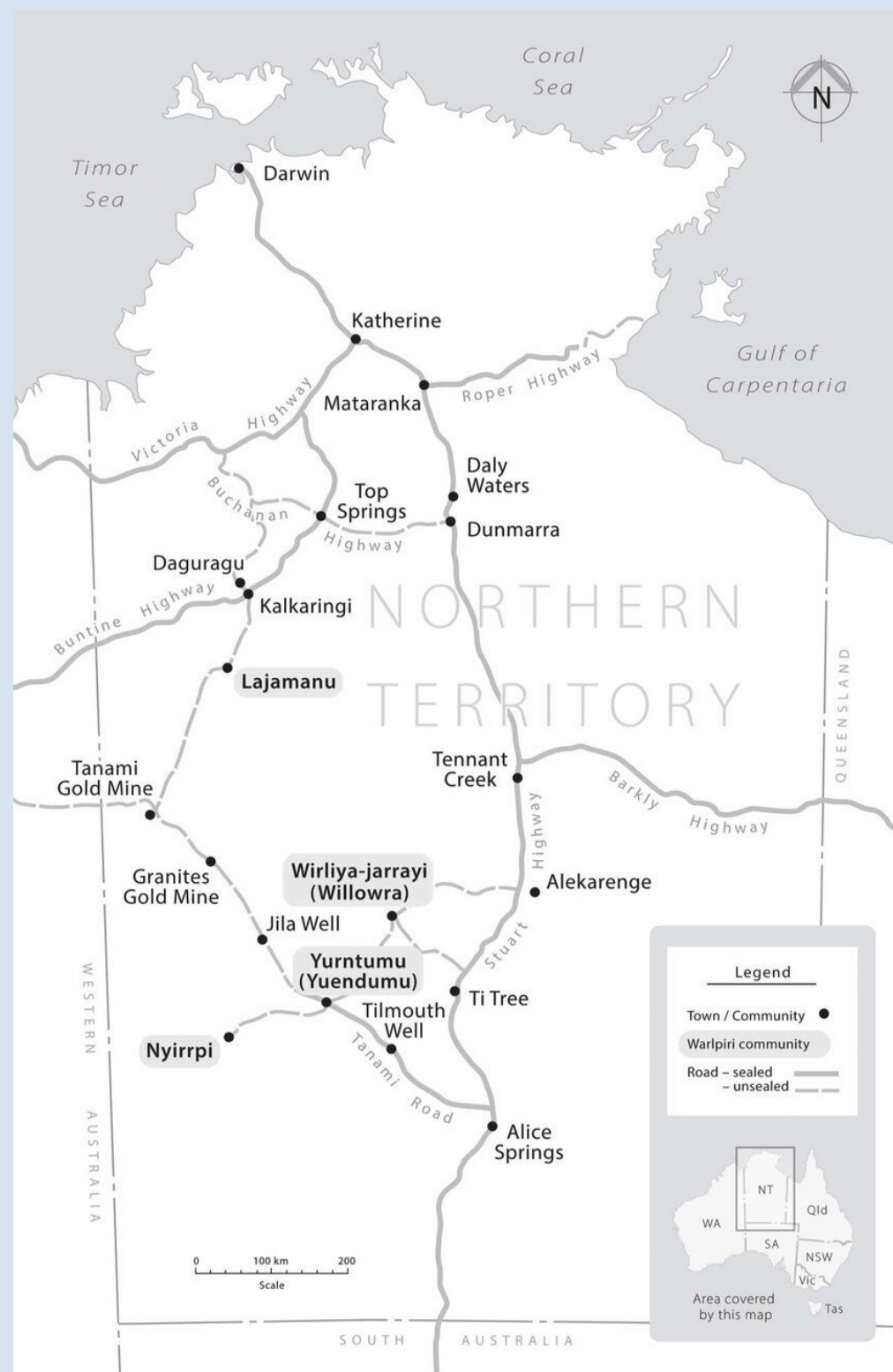
Warlpiri (Pama-Nyungan) has 5 stop POAs

	<i>bilabial</i>	<i>alveolar</i>	<i>retroflex</i>	<i>palatal</i>	<i>velar</i>
<i>stops</i>	p	t	ʈ	c	k

+ /t/ neutralizes with /ʈ/ between vowels
(Bundgaard Nielsen & O'Shannessy, 2019)

Data: Monologic narratives from DoReCo
(O'Shannessy, 2024), forced aligned in MAUS
(Schiel, 1999)

- 14 female speakers; aged 25-50
- 5,081 CV & VC tokens from VCV sequences



How do listeners distinguish different places?

Many cues used to distinguish stop POA (Delattre *et al*, 1955; Sussman & Shore, 1996) cross-linguistically

- Cues intrinsic to the stop:** e.g. duration (Butcher, 2012), spectral moments (Tabain *et al*, 2013), E-H/M (Bundgaard-Nielsen *et al*, 2016)
- Cues extrinsic to the stop:** e.g. formant transitions in adjacent Vs (Butcher 2012; Bundgaard-Nielsen *et al* 2012), vowel quality differences (Gallagher 2016)

QUESTION: DO VOWEL QUALITY DIFFERENCES (ALLOPHONY) HELP DISTINGUISH CONSONANTS MORE THAN TRANSITIONS?

OUR APPROACH:

- use maximum-likelihood classifiers to model optimal listener behaviour
- compare classifier performance with and without access to vowel quality

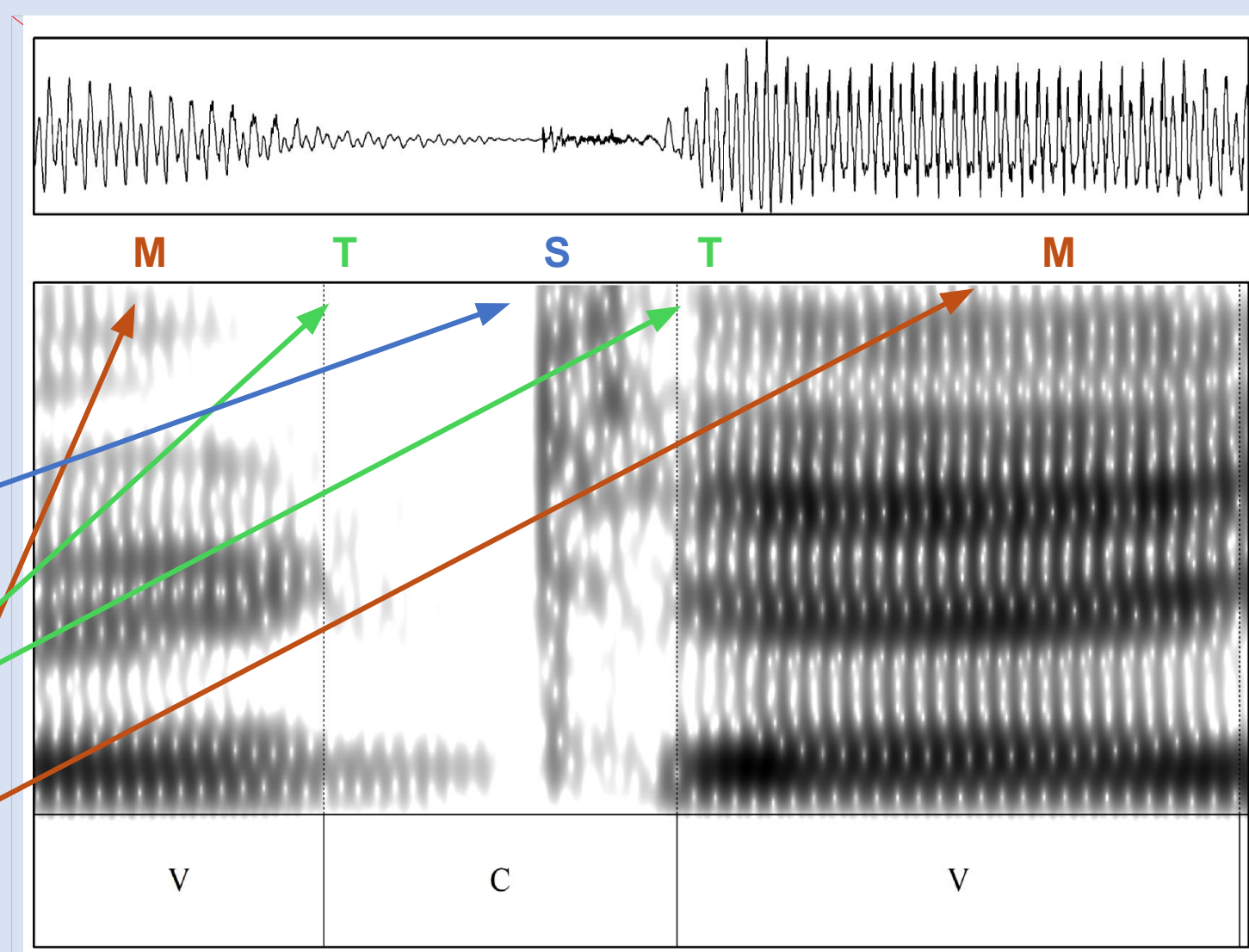
Maximum-Likelihood Classifiers

Phonetic tokens classified as category with highest likelihood of producing acoustic cue values of that token

- Simplifies Bayesian optimal listener models (e.g. Clayards *et al*, 2008; Feldman *et al*, 2009)
- assumes equal prior probabilities & perfectly sharp perceptual boundaries

Likelihoods from multivariate normal distributions over different cue set combinations fit to training data

- S:** stop-intrinsic cues
- T:** formant transition cues
- M:** formants of V midpoints

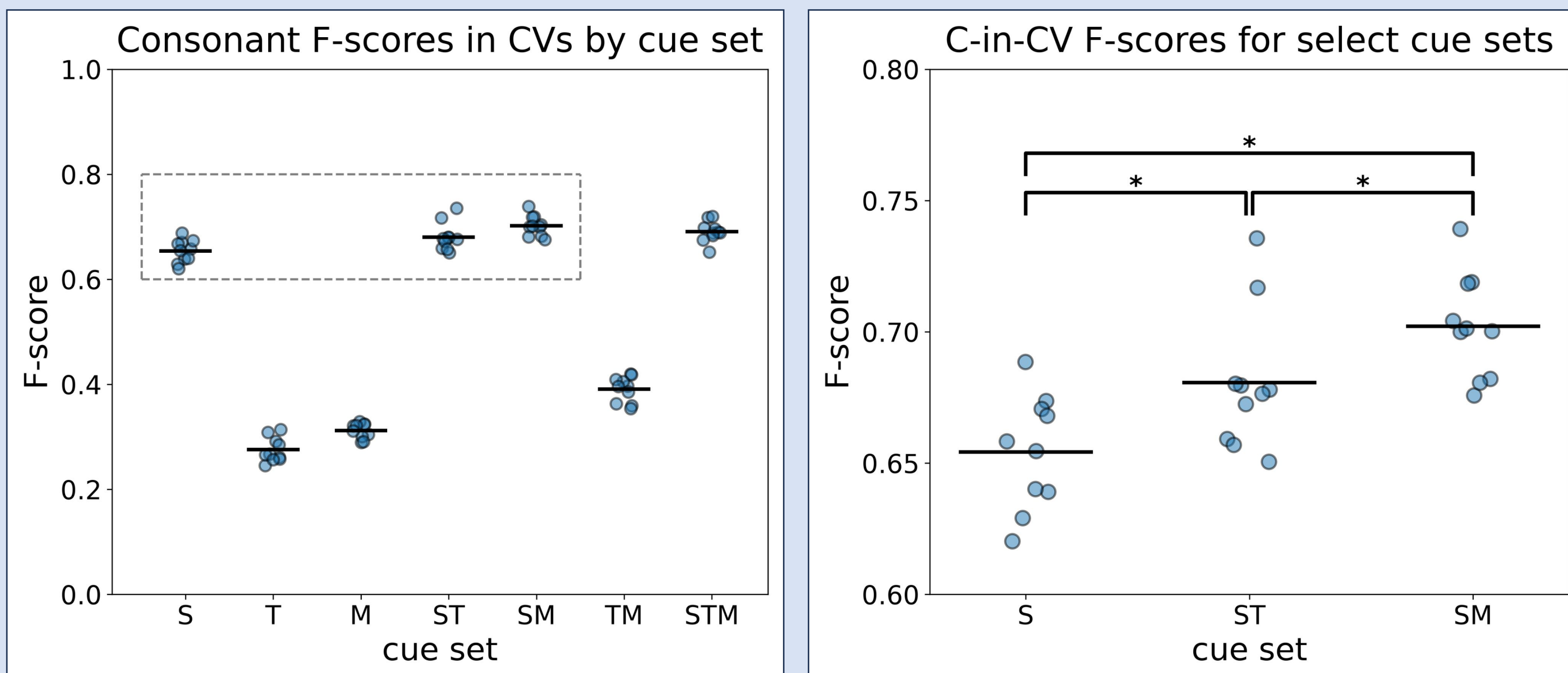


Model assessment method: Macro-averaged F-scores from stratified 10-fold cross-validation (Jurafsky & Martin 2026)

$\bar{d}^{(SM,S)}$ denotes improvement in average macro-F-score across folds for model with both S(top intrinsic cues) & M(idpoint) cues vs. just S cues

Permutation test (Menke & Martinez 2004) for significance testing

CV results: M(idpoint) cues help most



Consonant classification improves when T(ransition) cues added to S(top-intrinsic) cues

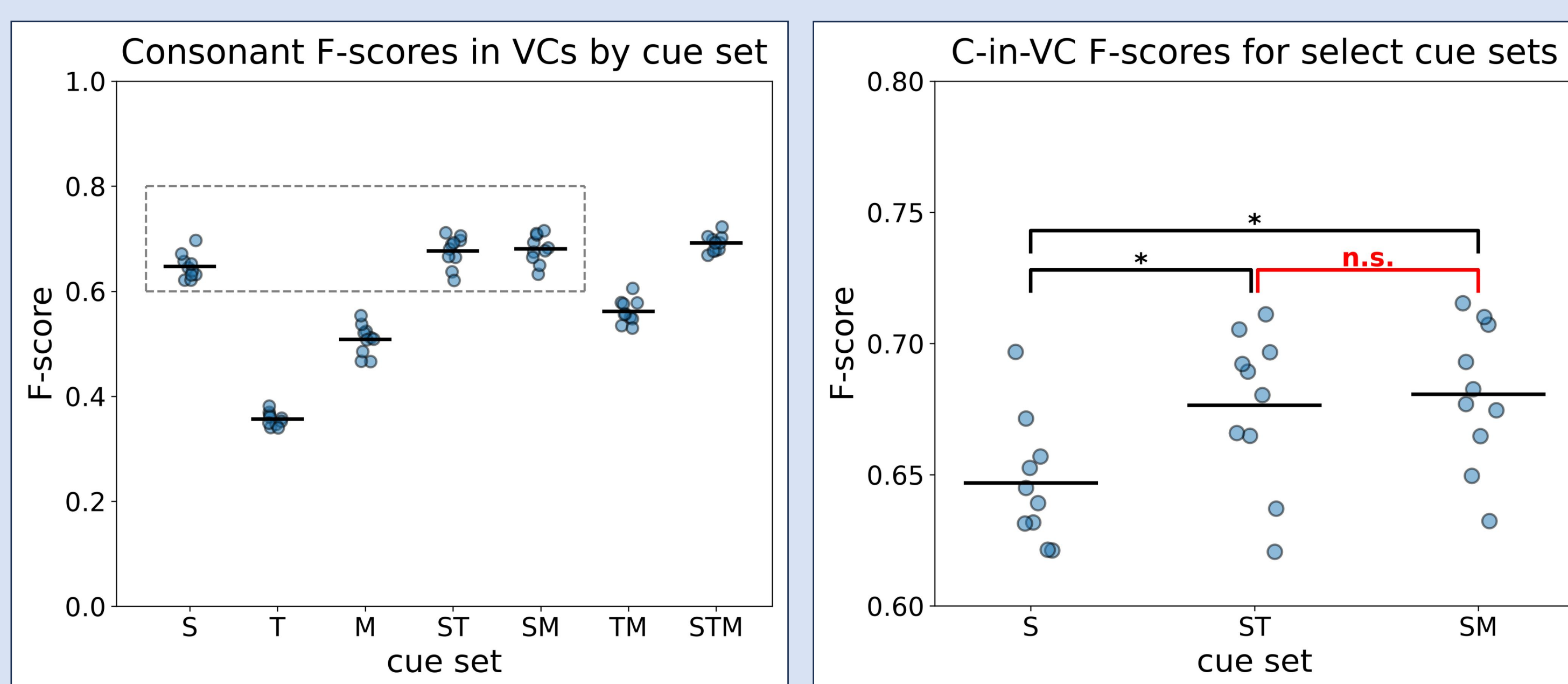
$\bar{d}^{(ST,S)} = 0.026, p < 0.005$

Improvement is higher with M(idpoint) cues

$\bar{d}^{(SM,S)} = 0.048, p < 0.005$

improvement doubles compared to T set: $\bar{d}^{(SM,ST)} = 0.021, p < 0.005$

VC results: similar pattern as CVs



Consonant classification improves when T(ransition) or M(idpoint) cues added to S(top-intrinsic) cues

$\bar{d}^{(ST,S)} = 0.030, p < 0.005$

$\bar{d}^{(SM,S)} = 0.034, p < 0.005$

Improvement is higher with M than T cues, but not significantly

$\bar{d}^{(SM,ST)} = 0.004, p = 0.140$

Difference from CV results perhaps due to stronger phonological association of CVs vs. VCs

Conclusions

Main takeaways:

Transitions & vowel midpoints improve classification of stop place: *vowel allophony helps C identification*

Ongoing work:

Analysis of specific cues within the S, T, and M cue sets for more fine-grained assessment of which cues are most important

Inventory size impact on allophony (McGahay 2026; Cram *et al*. 2026)

Meta-study and corpus analysis of acoustic structure of Australian phonetic space